

Project Design Phase: Solution Architecture

Date	19 September 2026
Team ID	
Project Name	ShopEZ — Apparel E-Commerce Application
Maximum Marks	4 Marks

1. Solution Architecture Overview

Solution architecture bridges business requirements and technical implementation. Its core objectives for ShopEZ include:

- Identifying the technical solution best suited to the identified business problem: accurate variant-level stock and unified admin management.
- Describing the software's structure, characteristics, and behavior for anyone reviewing the project.
- Defining the development phases and specific solution requirements that guided the build.
- Establishing a clear specification for how the solution is managed and delivered.

2. Proposed Solution Architecture Diagram

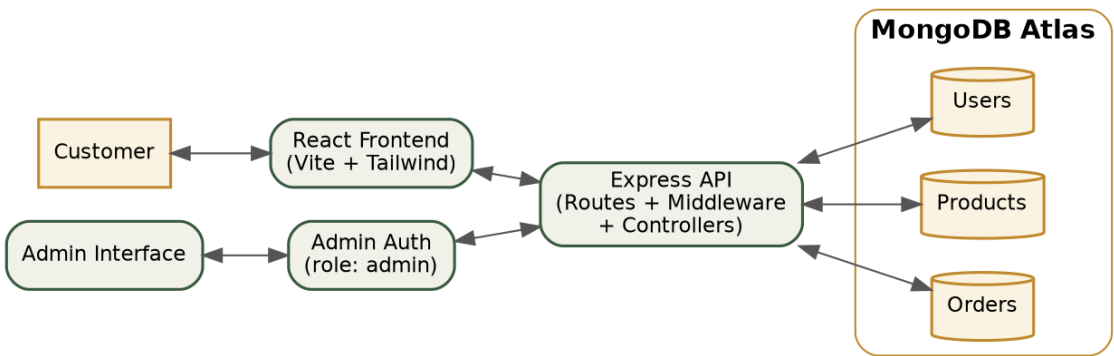


Figure 1: High-level MERN stack architecture flow for the ShopEZ platform

The customer interacts with the React frontend, which communicates with the Express API. The API reads from and writes to three core MongoDB collections — Users, Products, and Orders — shown here as the primary data the storefront depends on (Carts and Reviews follow the same pattern). The admin interface authenticates separately through the same API, gated by a role check, before reaching the same underlying data.